

Installation of Wireshark on Ubuntu

Before installing Wireshark, update the system using the following command on the Linux terminal:

```
$ sudo apt update
```

After this is done successfully, install Wireshark on Ubuntu by using the command given below:

```
$ sudo apt-get install wireshark
```

Next, launch Wireshark from the terminal by running the following command:

```
$ sudo wireshark &
```

Figure 2 shows the Wireshark

window when we run the above command on the terminal.

Mininet: Emulator for SDN

Mininet is an open source network emulator used to emulate SDN and its compatible components. It makes network devices such as the router and switches completely programmable, allowing the network to operate as per the user's needs. The main feature that discriminates SDN from a traditional network is the ability to customise protocols and features. Mininet supports the OpenFlow protocol, which is used to control the packet flow as per the API written on the SDN controller. It allows the creation of various nodes (controllers, switches, or hosts), enabling a huge network to be simulated on a single system. It

supports various customised topologies and guarantees their availability. The command line interface (CLI) and API make it easy to communicate with the network. Virtual hosts, switches and controllers are created using the Mininet emulator rather than hardware, and their behaviour is similar to distinct hardware.

Installation of Mininet

To install Mininet, run the command given below on the terminal:

```
# sudo apt-get install mininet
```

To confirm the successful installation of Mininet, use the following command:

```
# sudo mn
```

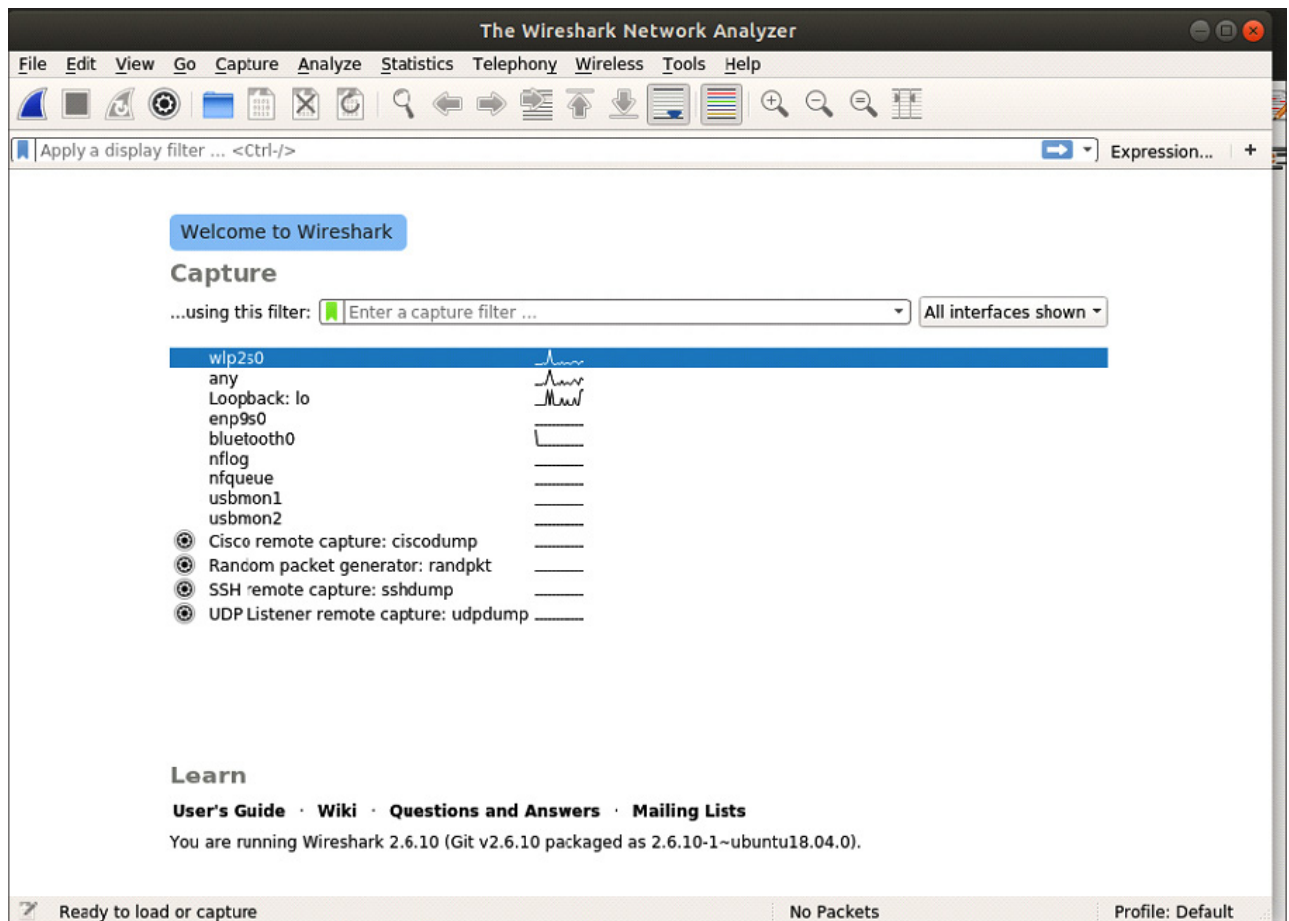


Figure 2: Wireshark window